When Assurance Work Displaces the Task

A Retrospective Case Study of Overdefensiveness and Overengineering in AI Agent Use

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Abstract

AI agents can impede work by adding confirmation, qualification, verification, and infrastructure, even when those activities appear individually reasonable. This paper examines how such work becomes disproportionate in one participant's sustained use of Codex. A retrospective, AI-assisted analysis revisits 32 episodes from 20 conversations selected from local records spanning April-November 2026, and adds three contrast episodes from three other conversations. The unit of analysis is a task-and-correction episode, not a model run or an independent failure. Overdefensiveness concerns risk handling beyond the demands of the task and its actual authorization; overengineering concerns complexity without a sufficiently demonstrated use. The trajectories show that process outputs can displace requested deliverables, extensive checking can miss the actual operating condition, and attempts to correct drift can themselves create further work. They also show that users sometimes authorize substantial assurance work and later reconsider it. Re-reading those authorizations changes the interpretation of apparent excess. The clearest local improvements involve recognizing existing capabilities, removing low-value supporting material, and aligning acceptance with the actual use path. Necessary safeguards and required independent review remain legitimate. The contribution is a task-relative account of assurance work, a deidentified episode catalogue, and propositions for subsequent evaluation. The evidence does not estimate prevalence, establish a treatment effect of personal instructions, or identify a model or harness as the principal cause.

Keywords: AI agents; human-AI interaction; overengineering; overdefensiveness; verification; qualitative case study

1. Introduction

Delegating a task to an AI agent also delegates some decisions about how much work to do. An agent may decide to request clarification, inspect a repository, introduce a reusable abstraction, construct a check, seek another review, or continue improving an already usable result. These decisions influence whether delegation reduces the user's work. A correct individual action can still contribute to an unsuitable overall process: a valid check may examine the wrong condition, and a carefully maintained record may be irrelevant to the requested deliverable.

This paper studies that problem through one user's sustained work with Codex. The records cover research writing, software maintenance, automation, knowledge management, and product exploration. The participant repeatedly objected to excessive caution and unnecessary complexity, introduced personal instructions, requested reviews and realistic task evaluations, and asked for specific simplifications. The records contain corrections that visibly changed local work, later complaints about similar behavior, and occasions where reducing procedure would have removed something the user actually required.

The question is therefore not whether agents should always do less. In one episode, the agent checked the appearance of current information after manual refreshes while the requirement concerned automatic updates. Reducing all verification would have missed the same requirement more cheaply. In another, the agent reduced an expressly required independent review to author self-checking. Here, doing less violated the task. Conversely, finding already implemented capabilities made proposed development unnecessary. These

distinctions require attention to what the work was for, who authorized it, and what evidence of its benefit exists.

I use assurance work to denote activities intended to make a result more trustworthy, maintainable, compliant with requirements, or safe to deliver. The term includes verification, permission handling, provenance records, review, qualification, and infrastructure built to support those activities. It is descriptive rather than pejorative. Some assurance work is essential; some is useful only in particular settings; some becomes a substitute for the result it was meant to support.

Three research questions organize the analysis. RQ1: How can disproportionate assurance and complexity be distinguished from justified safeguards and necessary task structure? RQ2: Through what observable sequences do process activities displace the requested outcome, and how do corrections change those sequences? RQ3: What do the records support about local improvement, recurrence, and attribution to models, instructions, and execution environments?

The paper contributes a task-relative distinction between overdefensiveness and overengineering; a process account grounded in reconstructed task-and-correction trajectories; and a public catalogue of deidentified episodes, including counterexamples and uncertain outcomes. It is a retrospective study of naturally accumulated records, not a prospective longitudinal experiment. The participant is also the author and the initiator of the investigation. AI assistance and that dual role are central limitations of the evidence, not independent validation of it.

2. Concepts and related work

2.1 Two distinct judgments

Overdefensiveness refers here to confirmation, qualification, protection, or error handling that exceeds the demands of the current task, existing authorization, and concrete risk. Examples include reopening a settled permission question without a changed consequence, or making an additional eligibility condition a prerequisite for an already authorized activity. The relevant comparison is not an unrestricted agent: it is an agent that respects actual permissions and explains uncertainty at the point where it affects a decision.

Overengineering refers to structure, functionality, workflow, or verification whose additional cost is not adequately connected to the current deliverable, an existing contract, or an observed failure. More files, tests, or agents are not sufficient evidence. A large requested study may need many of them. Evidence of disproportion is stronger when an addition duplicates existing capability, changes the deliverable, imposes a fixed obligation across unrelated tasks, or continues after the need that justified it has ended.

The concepts can overlap without being equivalent. Repeated confirmation can impede a task without adding software architecture. Redundant implementation can occur without caution or risk language. They interact when concern leads to another control, the control creates additional states or artifacts, and those artifacts invite further checking. The interaction is a proposed explanation of observable work, not a claim about a model's internal motivation.

Several neighboring failures must remain separate: incorrect status, premature stopping, unsupported claims of benefit, omitted required review, and ordinary weak analysis. Those failures can coexist with excessive work. They should not become overengineering merely because a dissatisfied user used that word. The catalogue retains such episodes because they help identify the boundaries of the explanation.

2.2 Instruction following, code quality, and interaction

Repository context files offer a direct way to influence coding agents, but compliance with instructions and improved task outcomes are different questions. Gloaguen et al. [1] evaluate generated and developer-authored context files across coding settings. They report no general improvement in task success and increased average inference cost, while recognizing a role for non-standard project practices. This motivates examining what a persistent instruction changes in actual work rather than treating its installation as evidence of benefit.

Two recent benchmarks address different aspects of design quality. SlopCodeBench [2] studies repeated extension of agents' own solutions and reports increasing structural erosion and verbosity. Its quality guidance improves initial code characteristics without removing later degradation. CodeTaste [3] distinguishes executing a specified refactoring from identifying the refactoring choices made by human developers. The present study examines a neighboring decision: whether a workflow, abstraction, or further evaluation should be added at all. Its evidence includes research and communication tasks, so code-size metrics alone would not capture the observed burden.

Human-AI interaction guidance distinguishes efficient correction from global control [4]. That distinction matters here: editing an unsatisfactory answer, changing a personal instruction file, and redesigning an agent workflow are different interventions. A local correction can be successful even when a global behavioral change has not been established. This study extends that interaction question to agents that can create additional tools and process artifacts while responding to correction.

Perceived productivity is also an inadequate substitute for observed outcomes. Becker et al. [5] found a mismatch between developers' perceived benefit and measured completion time in a randomized study of a particular early-2025 coding setting. That finding does not estimate the effect of the tools studied here. It supports the narrower methodological reason to keep claimed usefulness, observable task results, and cost separate.

The research design draws on case-study guidance in software engineering [6], especially explicit units of analysis, links from evidence to interpretation, and consideration of alternative explanations. The present work uses a structured retrospective comparison of episodes. It does not claim theoretical saturation, a grounded-theory procedure, or the controls of an experiment. Related work was selected to situate the research questions; it is not a systematic review of all agent complaints or mitigation techniques.

3. Study design and evidence

3.1 Setting and corpus

The setting is one participant's use of a desktop and command-line coding agent for software and knowledge work. Conversations are primarily in Chinese. Tasks vary in size, duration, and consequences. The participant also develops some of the tools later evaluated in the records and actively changes instructions and workflows. This provides continuity and contextual knowledge while making the setting unusually intervention-rich and unsuitable as a representative user sample.

The initial local inventory comprised 1,226 unique session IDs and 1,146,456 log lines; 26 lines could not be parsed. The inventory includes primary conversations, subagents, approval-related sessions, and other sources. It is not the analytical sample. The earlier descriptive report identified 344 candidate conversations from inventory and database metadata. A recount for this manuscript finds 343 distinct conversation IDs in the saved, searchable message extracts: 315 from session-derived extracts and 28 from database recovery, with no overlap between those two sets. This manuscript reports the observed extract count rather than assuming every inventory candidate supplied usable messages. Neither count is a denominator for a failure rate.

The searchable records span April-November 2026. The 32 previously selected episodes occur between June and November and come from 20 conversations. Several episodes belong to one long-running project conversation. An episode can include multiple requests, responses, corrections, and follow-up observations; episodes are not independent replications. Session source labels also do not guarantee that every candidate was a spontaneously user-authored task: some contain scripted research or review assignments. Selection of the core episodes depended on their task context, not merely their session label.

3.2 Selection and reanalysis

The initial selection was problem-oriented. Chinese and English search cues covered excessive defense, engineering, testing, repeated confirmation, unnecessary complexity, and related complaints. Requests, agent responses, and subsequent results were then read in context. This preferentially retrieves visible friction and misses problems the user did not name. It also cannot estimate how often ordinary work succeeded.

For the manuscript, the 32 episode identifiers were retained to preserve continuity with the descriptive report, but their earlier labels were not treated as settled judgments. The reanalysis checked the cited message locations, read neighboring requests where authorization mattered, and directly inspected retained artifacts for the proposal review, the two maintenance tasks, and the memory comparison. A private mapping links episode aliases to the source conversation and message locations. The public catalogue contains paraphrased observations, corrections, outcome status, and interpretive limits.

Three additional contrast episodes were purposively selected outside the 20 core conversations. Candidate conversations were inspected without relying solely on explicit complaint hits; short scripted research assignments were excluded from this contrast selection. The selected episodes concern a response-only connection check, a constrained summary, and compression of a professional memo after a substantive assumption had been specified. They were chosen because the requested output and the resulting response can be compared directly. They are boundary probes, not matched controls or an unbiased sample of successful work. The expanded catalogue therefore contains 35 episodes from 23 conversations.

Two re-readings materially qualify the prior account. In C06, the user explicitly requested a small governance tool and discussed ablation before later questioning a proposed expansion into routine gates and broader testing. Its initial construction cannot be described as wholly unsolicited. In C19, the user first approved an anonymous comparison for personal reading, then explicitly changed to model review in separate contexts. The subsequent review proceeded. This episode supports adaptation to revised review requirements, not the claim that any request for human reading was inherently unnecessary. No aggregate count of agent failures is reported in this paper.

3.3 Analytical procedure

Each episode was organized around six fields: the requested result; the relevant authorization; the observable activity; its relationship to that result; the correction or change in direction; and the strongest outcome supported by the record. The working concepts in Section 2 guided comparison, while the authorization re-readings constrained their application. The analysis asked both whether an addition had a demonstrated use and whether removing it would violate an explicit requirement.

Evidence is distinguished by what it can support. Messages establish expressed requirements, proposals, and conversational sequence. A close-out message can establish that the agent reported a change, but by itself does not independently verify a repository or a deployed system. A retained artifact can substantiate particular contents or results. No historical execution was rerun for this manuscript, and no historical deployment statement is promoted to a claim about current production state. Where an outcome remained proposed, reported, or unresolved, the catalogue preserves that status.

Cross-episode comparison focused on changed deliverables, misaligned acceptance, additional permission or review requirements, fixed process obligations, direct subtraction, and unsupported effectiveness claims. Counterexamples were used to ask whether the same surface behavior could be justified by the task. The

analysis did not assign a severity score, infer intent from wording, or use the agent's admission of fault as a sufficient verdict. Shorter output and fewer tool calls were not automatically coded as better work.

The principal findings below use deidentified reconstructions rather than verbatim quotations. All reconstruction paragraphs are paraphrases of retained requests and responses; they are not transcripts. Task names, exact dates, private paths, account identifiers, repository names, and operational details unnecessary to the argument have been removed or generalized. Relative order and the distinction between proposal and completion are retained. The three contrast episodes are interpreted only for directly visible compliance; their factual quality or downstream acceptance is not assumed.

3.4 Researcher position

The author is the sole human participant, the requester of the original tasks, and the initiator of this investigation. Codex assisted retrieval, selection, analysis, drafting, and document production. The same broader class of system being studied therefore helped construct the account. Re-reading source messages and retained artifacts reduces some unsupported assertions but does not create analytical independence. No independent human coding, inter-rater agreement study, or external expert validation was performed. Reviews described inside historical episodes were part of those tasks; they were not independent reviews of this manuscript.

4. Findings

4.1 Process outputs can change the deliverable

In C01, the requested result was an integrated report that could be read and circulated as a whole. The delivery separated narrative text from source and claim registers, uncertainty material, a quality checklist, and completion notes. The user required the relevant material to be integrated into the final document and distinguished working files from public-facing deliverables. The issue was not that sources or uncertainty should disappear. The additional reader work arose because the package required reconstructing the report from process artifacts. The retained exchange supports a delivery mismatch and a rework request; it does not establish acceptance of the eventual text.

C02 shows a stronger change of object. A research task moved toward layered prompting, agent structures, and system design. The user redirected it toward retrieval, analysis, and report writing. The same conversation also contains explicit requests for extensive coverage and repeated searches. Those requested activities must not be retrospectively counted as excess. The defensible distinction is between the authorized breadth of the research and an implementation structure that displaced the intended research product.

C08 and C25 extend the pattern in opposite directions. In C08, earlier work was carried forward as if it remained part of the current assignment, adding communication and application material to a repository task. In C25, a broad archiving request was narrowed to a preferred decision-tracing concept. An agent can impose its own conception of value by expanding or shrinking the task. Calling both errors excessive complexity would conceal the second mechanism: insufficient scope can also result from an overconfident procedural abstraction.

The common observation is a substitution of the object of completion. The agent can point to real artifacts, yet the user still lacks the specified result. The warranted remedy is to restore the deliverable and its reading or use path. It is not a universal ban on source registers, intermediate notes, or structured methods.

4.2 Extensive verification can miss the decisive condition

In C05, an information site was meant to update automatically on a schedule. Repeated checks after manual refreshes showed fresh data and a functioning page. The user objected that these observations did not establish unattended execution and asked for a near-term scheduled run. The agent then distinguished current freshness from automatic operation and described a test using the real scheduler without a manual invocation.

The reconstruction has four parts: a recurring automatic result was required; a manual route produced a current result; checks of that result were treated as progress toward automatic reliability; the correction moved acceptance to the real trigger. The available excerpt establishes the diagnosis and a revised validation plan. It is not used here as proof that the automation subsequently ran reliably over an extended period. C03 contains related concerns but is not an independent replication because task types and surrounding conditions differ.

C21 supplies stronger artifact evidence for the difference between ordinary acceptance and a verification-record tool's incremental benefit. In two retained maintenance tasks, affected existing tests passed both before and after the fixes. Examination of user-facing output exposed the defects: incomplete clickable text in one case and a progress display that did not reflect completed work in the other. Task-specific acceptance then checked the repaired behavior. The record tool stored valid results and allowed reuse after unrelated documentation changes, but the retained evaluation attributed no additional prevented delivery error to the tool in either task.

This does not show that the record tool is useless. It shows that correct recording and task-relevant detection are separable contributions. A check can be fresh and correctly recorded while addressing an insufficient acceptance condition. Likewise, the presence of many earlier checks does not imply that the missing scheduler check is unnecessary. The observed problem concerns relevance of evidence, not an optimum number of tests.

4.3 Correction can introduce new work, including work the user authorizes

C06 began with criticism of goal drift in a research task. Subsequent discussion explicitly included a small cross-task tool and compact personal instructions. The agent reported creating regression examples and configuration changes, then recommended a broader sequence involving real-task ablation, repeated review, receipts, and eventual routine gates. The user questioned whether those next steps were needed. The agent withdrew the broad expansion and proposed retaining narrower changes.

The initial tool was not an unauthorized act, and the user's earlier interest in ablation matters. The episode is better understood as a co-produced escalation whose expected usefulness was reconsidered. The agent's successful internal examples did not demonstrate improved outcomes in ordinary work. Nor did its later proposal to restore settings and remove the tool establish that removal had happened. The public catalogue therefore records a withdrawn recommendation and a proposed reduction, not a completed general cure.

C11 and C15 contain repeated objections to additional review, comparison, and small-patch cycles in a long project conversation. Some later iterations were explicitly requested or approved. The observation is that the interaction repeatedly returned to the question of whether further work served the current result. Counts of patches or releases would not resolve that question. The same participant sometimes asked for a specific comparison and elsewhere instructed the agent to stop extending the workflow.

These episodes complicate an account in which a model simply ignores a stable preference. An agent can influence what work is proposed; a user can authorize it; the work can then prove unhelpful or create further obligations. Authorization makes an action permissible, but does not establish that recommending it was useful. Conversely, a later preference for stopping cannot make all earlier work unauthorized. The temporal order is necessary to interpret both responsibility and benefit.

4.4 Concrete subtraction has more visible local outcomes than general rules

C10 concerns a proposed improvement to a research tool. At the user's request, a separate-context review inspected the proposal against existing implementation and retained diagnostic records. The review found that several proposed capabilities already existed, reduced two recommendations, and rejected dedicated integration code as unsupported by the current need. The retained review document supports this decision. The directly observed benefit was avoiding duplicate development, not a measured improvement in all subsequent research tasks.

C20 concerns low-value self-evaluation tasks and their supporting files. The user requested removing the recent additions while retaining useful browsing, consumption examples, complete cases, and substantive findings. Close-out messages report deleting the additional tasks, their dedicated export and checking material, and repeated qualifications, followed by commits and synchronization. Those messages substantiate the reported cleanup sequence; this manuscript does not independently rerun or recreate the historical cleanup. The intervention's scope is concrete enough to distinguish removal from a promise to behave differently.

Persistent personal instructions are a different kind of intervention. C14 records that personal working preferences had been placed in project contribution guidance and were later moved to personal global instructions. That change addressed a scope error. A later consolidation of the personal rules was also verified as saved and loaded during the original investigation. These observations concern deployment of instructions. The present corpus does not provide a controlled post-intervention period establishing their durable effectiveness.

Similar complaints recur across the retained record, but their recurrence is not a quantified treatment failure. Tasks, models, instructions, and tools changed; the user also changed requirements. The evidence supports local narrowing and an unresolved durability question. It does not support a percentage reduction in overengineering or a claim that a particular prompt reliably cures it.

4.5 Necessary safeguards and review delimit the claim

C27 concerns deletion of a source archive. A processed collection had been uploaded, but that did not establish a complete backup of the original directory. The user regarded further caution as excessive. The record nevertheless provides a task-specific reason for a coverage check: deletion could remove material absent from the processed collection. C32 concerns a credential that had previously appeared in a conversation. The proposed protective action addressed an observed exposure rather than an invented future risk. The specific credential and operational details are omitted from the public material.

C16 shows the opposite of excessive procedure. A required independent review was replaced by author self-checking in the name of simplification. The user explicitly rejected that substitution. Additional confusion about whether the host could run separate agents introduced an unnecessary route discussion. The episode combines omitted required work with avoidable friction; reducing everything would compound the problem.

C19 provides a useful authorization contrast. The user initially approved a blinded comparison prepared for personal reading. The user later requested separate-context model reviewers instead. The agent reused the existing outputs, obtained reviews, and reported substantive editing problems, including loss of useful judgments and narrative structure. This supports a local benefit from the revised review arrangement. It does not prove that human review was initially an invented prerequisite, that model review is generally equivalent to human judgment, or that the present paper has been independently validated.

The three added contrast episodes further limit an indiscriminate critique. In N01, the agent returned exactly the requested response to a connection check; the response itself did not verify system health. In N02, a numbered summary followed a requested count and sentence constraint; source fidelity was not re-evaluated. In N03, the agent compressed a professional memo after the user fixed a relevant assumption, retaining the requested sections and placeholders rather than reopening the entire analysis. These illustrate bounded execution visible in the record. They do not supply a general success rate.

5. A process account and alternative explanations

The trajectories suggest that assurance work becomes problematic when the object being optimized moves away from the required result. One possible sequence begins with uncertainty or dissatisfaction, proceeds to a proposed control or supporting artifact, and then evaluates the existence or internal correctness of that artifact. If that evidence is treated as sufficient progress, further work can center on maintaining the control rather than resolving the original need. A later complaint may then trigger another layer. Figure 1 summarizes this interpretation.

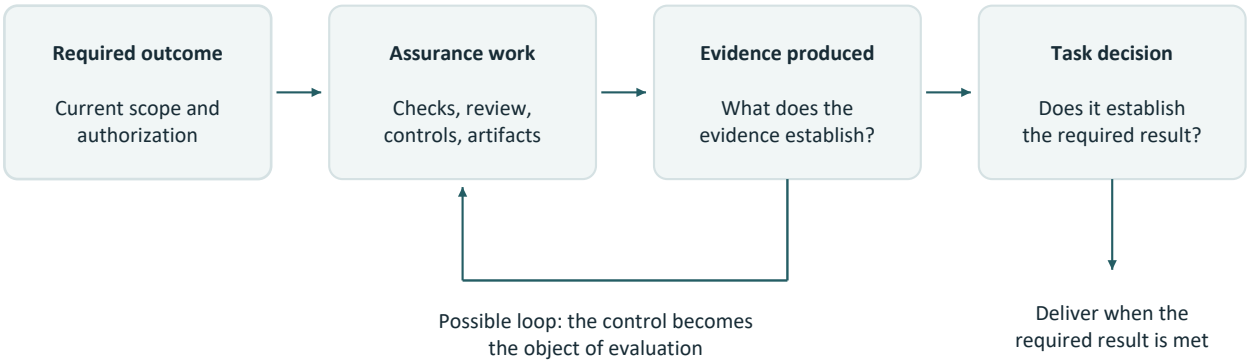


Figure 1. A task-relative account of assurance work. The loop is an interpretation of selected trajectories; it is not a measured causal relationship. Necessary controls remain part of the authorized task.

The sequence is most visible in the reconsidered governance proposal and in evaluations where mechanism checks preceded realistic task assessment. It is not established in every episode, and no hidden reasoning or training signal was observed. The figure is an interpretive model of the interaction, not a measured causal graph. The same added control can be appropriate when it addresses a demonstrated failure or required independent responsibility.

Three propositions follow for future study. P1: When completion is evaluated primarily through readily counted process outputs, an agent can satisfy those outputs while failing the actual use condition. P2: Corrections that identify a concrete redundant artifact, unsupported obligation, or missing acceptance condition are more directly auditable than broad instructions to be less defensive. P3: Prior authorization and subsequent renegotiation moderate judgments of excess; reconstructing their order reduces false attribution of unnecessary work to the agent alone. These are propositions generated by the cases, not tested causal effects.

Several rival explanations remain plausible. Ambiguous tasks may require additional clarification. Specific project rules or platform restrictions may force a step that appears unnecessary in a short excerpt. The user may change preferences after seeing an output. Environmental failures can lengthen work without a poor design decision. A missing source or unverified deployment can justify a qualification. Each of these possibilities weakens any inference made solely from a complaint, an apology, or a high tool-call count.

The strongest findings survive these alternatives at a narrower level. Discovering existing capability defeats the premise of duplicate development in C10. A manual refresh does not exercise an automatic trigger in C05. A valid record does not by itself demonstrate an incremental prevented error in C21. A source archive and a processed derivative do not establish the same recovery coverage in C27. These relationships are inspectable even when the model's internal cause and the user's general satisfaction remain unknown.

The corpus cannot separate model behavior from the execution environment. Turn-level labels in some retained records span multiple model generations, while database-recovered conversations may expose only current thread metadata. The harness determines available tools, injected instructions, memory, continuation behavior, and permission handling. Historical variation in those conditions is confounded with task difficulty

and user intervention. The evidence supports a problem that occurs in the combined interaction; it does not rank models or assign principal responsibility to the harness.

6. Implications for design and evaluation

The first implication is to preserve the requested object of completion. An agent's working representation should keep the deliverable, current scope, and already granted authorization available as the task evolves. This need not take the form of another mandatory document. In the observed research tasks, the useful correction was to produce the integrated report; an additional report about why integration matters would have added work without resolving the omission.

Second, additions should have a stated task-specific use. The relevant question is what existing requirement, contract, or observed failure the addition addresses. A prior approval answers whether the agent may proceed; it does not establish net value. A proposed abstraction that duplicates an existing function should be withdrawn. A requested independent review should remain. A credential protection step triggered by actual exposure should not disappear merely because the user prefers concise interaction.

Third, verification should exercise the condition the user will rely on. For scheduled work, that includes the trigger rather than only the resulting page. For a verification-record tool, evaluation should distinguish accurate record keeping from errors it prevents beyond ordinary acceptance. For writing, a change in length or a passed format check should not substitute for preservation of the argument and useful content. These are different outcome measures for different tasks.

Fourth, interventions should be described at the level demonstrated. Saving an instruction file demonstrates configuration deployment. Cancelling duplicate development demonstrates a local decision change. A committed cleanup is stronger evidence of removal than a recommendation to clean up. Sustained improvement requires observations across subsequent tasks under an interpretable comparison. This distinction avoids both celebrating an untested cure and dismissing a local improvement because it is not universal.

For a future intervention study, the smallest informative design would hold the model, harness, tools, task snapshot, and resource limits constant while varying the instruction or workflow under examination. Evaluation would jointly consider delivery quality, unnecessary work, user corrections, and cost. Failures and timeouts would remain in the analysis. C22 illustrates the need: all three retained memory-condition runs reached the time limit, so partial progress and retrieved requirements cannot establish a winner in total completion time. New experiments were not conducted for this paper.

These implications are not a proposal for a universal approval gate or an additional evaluation platform. They identify decisions that a designer or evaluator must make explicit when studying the problem. The present evidence cannot establish that mechanically enforcing these decisions would improve all tasks, or that reducing tests or review in general is beneficial.

7. Limitations, privacy, and reproducibility

This is a single-participant, problem-oriented retrospective study. The participant is highly engaged, repeatedly intervenes, and develops tools used in some episodes. Those characteristics can intensify both visibility of friction and opportunities for correction. The study is useful for examining processes in context, but not for estimating their distribution across ordinary users, tasks, or products.

Selection favors complaints and salient corrections. Three purposive contrasts cannot repair that bias or function as a control group. The core episodes were not prospectively sampled, and the analysis does not establish saturation. Records from the same conversation share history, instructions, and project state. The catalogue's 35 entries therefore represent analytical distinctions, not 35 independent observations of a statistical treatment.

Some claims rely on close-out messages rather than independently verified historical state. Direct inspection of retained evaluation results strengthens particular findings, but the artifacts were themselves produced in AI-assisted work. Agreement between an agent's narrative and its own record is not the same as an external audit. Missing log lines, source filtering, truncation of older records, and the mismatch between inventory candidates and extractable conversations further limit completeness. The study does not claim to have reconstructed every task the participant ever performed.

AI-assisted selection and interpretation can reproduce the very tendencies under investigation, including excessive qualification, post-hoc rationalization, and accepting the user's framing too readily. Re-reading authorization changed the interpretation of C06 and C19 and was therefore substantively useful, but it is not an independent reliability study. The findings are presented as a transparent exploratory account with explicit evidence levels. Independent reanalysis would strengthen the work.

The author is publicly identified, so the study is not anonymous at the participant level. Deidentification instead protects task-level and third-party details. Public materials omit raw conversations, private source paths, account details, repository and project identifiers, exact event times, credentials, and the private crosswalk. Reconstructions preserve only information needed for the analysis. Names and numbers that could unnecessarily reconnect cases to operational records are generalized. The three contrast episodes are described as interactions, without redistributing their source documents or professional subject matter.

The accompanying CSV exposes the observations, corrections, outcome status, and interpretive limits for every catalogue entry. The codebook defines these fields and the meaning of evidence levels. This permits scrutiny of the published account and its internal reasoning, but not independent reproduction of all coding from raw data. Raw logs and the crosswalk remain private; access to them is not promised. No claim of institutional ethics approval, independent human validation, or participant anonymity is made.

8. Conclusion

Overdefensiveness and overengineering concern different judgments: how much protection a task needs and how much complexity its result justifies. In this case, they sometimes interacted through additional checks, artifacts, and governance work, while important failures also arose from insufficient review or the wrong verification condition. The most informative corrections were concrete: restore the deliverable, inspect existing capability, remove a low-value addition, or test the actual use path. Their local effects should be distinguished from the unproven durability of general rules. Reconstructing authorization also matters: work can be requested and later reconsidered without having been unsolicited from the start. The evidence supports studying proportionality as a property of the evolving human-agent task, rather than inferring it from caution, complexity, or user dissatisfaction alone.

Declarations

Author and participant role. Song Luo supplied the task requirements and corrections in the underlying interactions, initiated the retrospective investigation, and authorized publication. The author has a direct interest in some tools discussed in deidentified form. This relationship may influence judgments of usefulness and burden.

AI assistance. Codex assisted source inspection, literature retrieval, episode organization, analysis, manuscript drafting, and preparation of the PDF and public catalogue. No AI system is listed as an author. Historical reviews performed by separate agents are described only as evidence within their respective episodes, not as peer review of this paper.

Data availability. The public episode catalogue and codebook accompany this preprint. Raw conversational logs, private artifacts, and the identifying source crosswalk are withheld. The public reconstructions are paraphrases, not verbatim transcripts. This preprint reports no new controlled experiment.

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Appendix A. Episode catalogue

The catalogue retains 32 core episodes and adds three purposive contrasts. The 35 entries belong to 23 conversations. Shared conversation aliases indicate dependence. Entries preserve local outcomes and limits; they are not a binary list of failures.

Evidence notation. M: messages and contextual reading. C: a contemporaneous close-out report of execution. A: a retained artifact inspected in this investigation. C remains an agent report; A is not necessarily independent evidence. The companion CSV provides the same entries, and the codebook describes their use.

C01 | T01 | Research report

Observation. Delivery fragmented into process files; integration requested.

Correction. Requested one readable final report.

Recorded outcome. Rework; acceptance unknown

Evidence: M: source messages and contextual reading
Limit: The rework request is visible; acceptance of the eventual report is not established.

C02 | T02 | Research task

Observation. Investigation became system design; original goal restored.

Correction. Removed or reduced system design; restored research and writing.

Recorded outcome. Direction corrected

Evidence: M: source messages and contextual reading
Limit: Extensive research was explicitly requested. The criticized substitution was system design, not research breadth.

C03 | T03 | Information site A

Observation. Extensive checks missed automatic operation.

Correction. Made automatic execution a distinct acceptance target.

Recorded outcome. Acceptance target needed correction

Evidence: M: source messages and contextual reading
Limit: Large check counts alone do not show unnecessary testing. The automatic-use condition is the relevant gap.

C04 | T03 | Site deployment

Observation. Executable work was returned to the user despite authorization.

Correction. Requested execution of configuration work the agent could perform.

Recorded outcome. Execution route corrected

Evidence: M: source messages and contextual reading
Limit: Authorization to perform the configuration was explicit. Actual login or verification challenges may still require user action.

C05 | T04 | Information site B

Observation. Manual refresh substituted for evidence of scheduled execution.

Correction. Requested an actual scheduled run shortly afterward.

Recorded outcome. Actual scheduled acceptance adopted

Evidence: M: source messages and contextual reading
Limit: The excerpt establishes a corrected validation target and plan, not sustained successful scheduled operation.

C06 | T05 | Goal governance

Observation. A requested small governance tool was followed by proposals for broader ablation, review, receipts, and routine gates.

Correction. The user questioned the next steps; the agent withdrew the broad expansion and proposed narrower changes.

Recorded outcome. Expansion withdrawn in conversation; proposed configuration restoration and tool removal not counted as completed.

Evidence: M: source messages and contextual reading
Limit: The user requested the initial small tool and discussed ablation. Later withdrawal is scope renegotiation, not proof that all construction was unsolicited.

C07 | T06 | Memory-tool architecture

Observation. External review questioned architectural complexity.

Correction. Moved toward real-task observation rather than judging by size.

Recorded outcome. No blanket judgment justified

Evidence: M: source messages and contextual reading
Limit: Architectural scale and an external critique do not establish disproportion relative to the full requirements.

C08 | T06 | Project improvement

Observation. Earlier tasks entered currently authorized work.

Correction. Restated current scope and withdrew extra work.

Recorded outcome. Acknowledged and narrowed

Evidence: M: source messages and contextual reading
Limit: The added material was not reported as sent or published. Scope drift is distinct from irreversible external action.

C09 | T07 | Research-tool review

Observation. Independent review became a requirement for outside people.

Correction. Restated authorization for review in a clean context.

Recorded outcome. Repeated correction and recurrence

Evidence: M: source messages and contextual reading
Limit: Separate-context review was explicitly permitted. This does not establish general equivalence of model and human review.

C10 | T08 | Research-tool proposal

Observation. Existing capabilities were mistaken for development gaps.

Correction. Reduced or cancelled proposals after separate-context read-only review.

Recorded outcome. Two recommendations reduced; dedicated integration code rejected; no product code changed in this review.

Evidence: M+A: source messages and retained separate-context proposal review
Limit: The retained review supports avoiding duplicate development. No general task-quality or cost advantage was measured.

C11 | T09 | Research execution

Observation. Confirmation, checking, and review accumulated.

Correction. Specified necessary verification and stopping conditions.

Recorded outcome. Local reduction and recurrence

Evidence: M: source messages and contextual reading
Limit: Some reviews and repetitions were requested. Interpret only the identified additional obligations in their local authorization context.

C12 | T09 | Version comparison

Observation. Targeted benefit was absent and the newer version was slower.

Correction. Retained initial outputs, timing, and no-gain results; stopped expansion.

Recorded outcome. No-gain result retained

Evidence: M: source messages and contextual reading
Limit: The comparisons were authorized. A no-gain result is an empirical outcome, not evidence that conducting the comparison was itself excess.

C13 | T09 | Modes and quotas

Observation. Fixed modes and recording quotas added burden.

Correction. Removed fixed modes and quotas.

Recorded outcome. Product simplification initiated

Evidence: M: source messages and contextual reading
Limit: Fixed modes and quotas were reconsidered against the intended product. Their removal does not demonstrate improved research quality.

C14 | T09 | Personal-rule deployment

Observation. Personal preferences entered project contribution rules.

Correction. Withdrew project changes and saved personal global instructions.

Recorded outcome. Withdrawn and stored globally

Evidence: M: source messages and contextual reading
Limit: Saving and loading personal instructions establishes deployment, not a durable behavioral treatment effect.

C15 | T09 | Repeated small changes

Observation. Fixes and comparisons generated further tasks.

Correction. Removed fixed loops and requested an end to automatic iteration.

Recorded outcome. Fixed loops removed; work stopped

Evidence: M: source messages and contextual reading
Limit: Some later patches and releases were separately authorized. The episode concerns automatic continuation and repeated scope negotiation.

C16 | T09 | Review and agent access

Observation. Independence weakened while unnecessary access conditions were added.

Correction. Restored independent review using existing host capabilities.

Recorded outcome. Requirements and route corrected

Evidence: M: source messages and contextual reading
Limit: Required independent review must be preserved. Omitting it is under-delivery even if motivated by simplification.

C17 | T09 | Later research-tool responses

Observation. The user objected to a heavily qualified retrospective answer and requested actual use of a named retrieval capability.

Correction. Requested an actual attempt before judging feasibility.

Recorded outcome. Request for a direct attempt and clearer judgment is visible; no durable effect established.

Evidence: M: source messages and contextual reading
Limit: Dissatisfaction with caveats is a signal to inspect, not sufficient proof that every stated limit was unnecessary.

C18 | T10 | Prompts for separate products

Observation. Roles duplicated and product scopes mixed.

Correction. Deduplicated roles and separated instructions by product.

Recorded outcome. Roles deduplicated; instructions separated

Evidence: M: source messages and contextual reading
Limit: The user retained the role structure and board. The correction concerns overlap and scope, not eliminating collaboration.

C19 | T11 | Full-text editing review

Observation. An anonymous editing comparison was prepared for personal reading; the user then requested model review in separate contexts.

Correction. Reused original outputs and switched to the newly authorized review arrangement.

Recorded outcome. The reviews identified loss of useful judgments and disrupted narrative structure; local revisions were described.

Evidence: M: source messages and contextual reading
Limit: The user first approved an anonymous package for personal reading, then requested model review. Treat this as revised authorization, not an initially invented human gate.

C20 | T12 | Data and research assets

Observation. Low-value self-evaluation generated supporting machinery.

Correction. Removed low-value supporting material and retained useful content.

Recorded outcome. Removal of low-value tasks, dedicated supporting files, and repeated qualifications reported as committed and synchronized.

Evidence: M+C: source messages and historical cleanup close-out
Limit: Cleanup is supported by contemporaneous close-out reports. This manuscript did not independently reconstruct the historical commits.

C21 | T13 | Verification-record tool

Observation. Mechanism checks were treated as practical value.

Correction. Narrowed claims and simplified use after real repair tasks.

Recorded outcome. Actual output acceptance exposed defects; valid receipts were retained; no additional prevented delivery error observed.

Evidence: M+A: source messages and retained two-task evaluation results
Limit: Two tasks demonstrate mechanism operation, with no additional prevented delivery error observed. They do not establish absence of value in other settings.

C22 | T14 | Memory-tool reassessment

Observation. Size drove an overly strong complexity judgment.

Correction. Compared no memory, ordinary notes, and system memory.

Recorded outcome. All three conditions reached the same time budget; no completed-duration ranking established.

Evidence: M+A: source messages and retained three-condition execution results
Limit: All three runs timed out. Partial progress and retrieval success cannot establish a total-duration winner.

C23 | T14 | Repair delivery

Observation. An intermediate artifact became the stopping point.

Correction. The user again requested completion of remaining work.

Recorded outcome. Merge completed after further prompting

Evidence: M+C: source messages and historical merge close-out
Limit: A pull request was an intermediate result relative to the requested endpoint. Historical merging is reported, not reverified as current deployment.

C24 | T15 | API execution route

Observation. Confused product mechanisms complicated a simple request.

Correction. Clarified invocation and continuation, restoring a viable route.

Recorded outcome. Host and invocation roles clarified

Evidence: M: source messages and contextual reading
Limit: The interaction concerns a mistaken interpretation of the requested execution route; it is not a current product or billing guide.

C25 | T16 | Archive scope

Observation. Full archiving narrowed to a subset of traces.

Correction. Restored the complete archive and synchronization scope.

Recorded outcome. Original scope restored

Evidence: M: source messages and contextual reading
Limit: Reducing scope can also violate a task. The original broad archive requirement must be preserved.

C26 | T16 | Synchronization status

Observation. Completion overstated, followed by repeated qualifications.

Correction. Distinguished completed and pending steps directly.

Recorded outcome. Status and wording corrected

Evidence: M: source messages and contextual reading
Limit: Upload and device download are distinct states. Communication failure does not justify asserting a state not established by evidence.

C27 | T16 | Backup before deletion

Observation. A processed archive did not establish complete raw backup.

Correction. Explain coverage and gaps; this report retains the check.

Recorded outcome. Coverage check retained

Evidence: M: source messages and contextual reading
Limit: The processed archive was not demonstrated to be a complete raw backup. The protective check has a concrete recovery purpose.

C28 | T17 | Industry-trend writing

Observation. The user rejected a trend report for weak synthesis, forced grouping, and repeated qualifications.

Correction. The user stopped rewriting; this is not successful remediation.

Recorded outcome. User stopped the rewriting; requirements were not met.

Evidence: M: source messages and contextual reading
Limit: Poor synthesis and repetitive qualification coexist. Necessary distinctions about availability should not automatically be removed.

C29 | T18 | Research communication

Observation. Boundary discussion preceded the substantive explanation.

Correction. Strengthened the research explanation and revised the presentation.

Recorded outcome. Explanation strengthened and text revised

Evidence: M: source messages and contextual reading
Limit: The visible revision responds to communication needs; it does not show that all uncertainty statements are unjustified.

C30 | T19 | Portable terminal A

Observation. Feasibility discussion expanded prematurely into integration design.

Correction. Prioritized feasibility and routes consistent with stated preferences.

Recorded outcome. Answer and proposal narrowed

Evidence: M: source messages and contextual reading
Limit: The user later requested implementation planning. Only the earlier exploratory stage is criticized for unsuitable detail.

C31 | T20 | Portable terminal B

Observation. Complex options obscured the feasibility answer.

Correction. Separated the feasibility answer from implementation routes.

Recorded outcome. Direct feasibility and route requested

Evidence: M: source messages and contextual reading
Limit: The record supports a request for a more direct feasibility answer, not a completed implementation or general feasibility conclusion.

C32 | T04 | Credential protection

Observation. A previously exposed credential justified replacement.

Correction. Retain justified credential replacement while simplifying secure handling.

Recorded outcome. Safeguard retained; workflow could improve

Evidence: M: source messages and contextual reading
Limit: The prior exposure supplies a concrete basis for protection. No credential or infrastructure detail is published.

N01 | T21 | Response-only connection check

Observation. The user requested one exact response and prohibited tools and file operations. The reply matched the requested sentence.

Correction. No corrective turn in the retained exchange.

Recorded outcome. Visible response-format compliance.

Evidence: M: two-message exchange
Limit: The sentence is an instructed acknowledgement, not independent evidence that the system was healthy.

N02 | T22 | Constrained summary after extraction

Observation. The user requested ten numbered insights, each one or two sentences. The agent returned a ten-item summary.

Correction. Followed the newly specified summary format.

Recorded outcome. Visible count and format compliance.

Evidence: M: request and complete summary response
Limit: The underlying source fidelity and usefulness were not independently re-evaluated; absence of a complaint does not establish acceptance.

N03 | T23 | Compression of a professional memo

Observation. After a relevant business assumption was specified, the user requested a shorter memo with three sections and placeholders.

Correction. Compressed the memo while preserving the specified sections and placeholders.

Recorded outcome. The requested shortened text was returned without reopening the whole analysis.

Evidence: M: sequence of requests and complete final response
Limit: This is evidence about interaction scope and format, not verification of the professional advice or its real-world use.